

Menu-Driven C Program: Even and Odd Digits

1. Problem Understanding

Write a menu-driven C program using **switch-case** that accepts a positive integer and performs one of these operations:

1. Count the number of **even and odd digits**.
2. Find the **sum of even digits and sum of odd digits separately**.

Examples

For 12045:

- Even digits: 2, 0, 4 → **Even_dig = 3**
- Odd digits: 1, 5 → **Odd_dig = 2**

For 12345:

- Even digits: 2, 4 → **Sum_even = 6**
 - Odd digits: 1, 3, 5 → **Sum_odd = 9**
-

2. Algorithm

Operation 1: Count Even and Odd Digits

1. Start.
2. Input a positive integer num.
3. Initialize even_dig = 0 and odd_dig = 0.
4. Store num in temp.
5. Extract the last digit using digit = temp % 10.
6. Check whether digit % 2 == 0.
7. If true, increment even_dig; otherwise increment odd_dig.
8. Remove the last digit using temp = temp / 10.
9. Repeat steps 5–8 until temp becomes 0.
10. Display even_dig and odd_dig.
11. Stop.

Operation 2: Sum Even and Odd Digits

1. Start.
2. Input a positive integer num.
3. Initialize sum_even = 0 and sum_odd = 0.
4. Store num in temp.

5. Extract the last digit using $\text{digit} = \text{temp} \% 10$.
6. If the digit is even, add it to sum_even .
7. Otherwise, add it to sum_odd .
8. Remove the last digit using $\text{temp} = \text{temp} / 10$.
9. Repeat until all digits are processed.
10. Display sum_even and sum_odd .
11. Stop.

3. Test Case Table

TC Input		Number Choice	Expected Output	Purpose
1	12045	1	Even_dig = 3, Odd_dig = 2	Given example
2	12345	2	Sum_even = 6, Sum_odd = 9	Given example
3	2468	1	Even_dig = 4, Odd_dig = 0	All digits even
4	13579	1	Even_dig = 0, Odd_dig = 5	All digits odd
5	2468	2	Sum_even = 20, Sum_odd = 0	All digits even
6	13579	2	Sum_even = 0, Sum_odd = 25	All digits odd
7	10020	1	Even_dig = 4, Odd_dig = 1	Tests zeros
8	10020	2	Sum_even = 2, Sum_odd = 1	Tests zeros
9	12345	3	Invalid choice	Invalid menu choice
10	0	1	Invalid input	Non-positive input

4. C Program

```
#include <stdio.h>
```

```
int main(void)
```

```
{
```

```
    long long num, temp, digit;
```

```
    int choice;
```

```
    int even_dig, odd_dig;
```

```
    int sum_even, sum_odd;
```

```
printf("Enter a positive integer: ");
```

```
if (scanf("%lld", &num) != 1 || num <= 0)
```

```
{
```

```
    printf("Invalid input! Please enter a positive integer.\n");
```

```
    return 0;
```

```
}
```

```
printf("\n--- MENU ---\n");
```

```
printf("1. Count even and odd digits\n");
```

```
printf("2. Sum of even and odd digits separately\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch (choice)
```

```
{
```

```
    case 1:
```

```
        even_dig = 0;
```

```
        odd_dig = 0;
```

```
        temp = num;
```

```
        while (temp > 0)
```

```
        {
```

```
            digit = temp % 10;
```

```
            if (digit % 2 == 0)
```

```
            {
```

```
                even_dig++;
```

```
            }
```

```
            else
```

```
            {
```

```
        odd_dig++;  
    }  
  
    temp = temp / 10;  
}  
  
printf("\nEven_dig = %d\n", even_dig);  
printf("Odd_dig = %d\n", odd_dig);  
  
break;
```

case 2:

```
    sum_even = 0;  
    sum_odd = 0;  
    temp = num;  
  
    while (temp > 0)  
    {  
        digit = temp % 10;  
  
        if (digit % 2 == 0)  
        {  
            sum_even += digit;  
        }  
        else  
        {  
            sum_odd += digit;  
        }  
  
        temp = temp / 10;  
    }
```

```
printf("\nSum_even = %d\n", sum_even);
printf("Sum_odd = %d\n", sum_odd);

break;

default:
    printf("\nInvalid choice! Please choose 1 or 2.\n");
}

return 0;
}
```

5. Compilation

Save the program as:

even_odd_digits.c

Compile using GCC:

cc even_odd_digits.c

Run the program:

./a.out

6. Execution and Testing

Test Case 1

Input:

Enter a positive integer: 12045

--- MENU ---

1. Count even and odd digits
2. Sum of even and odd digits separately

Enter your choice: 1

Output:

Even_dig = 3

Odd_dig = 2

Result: PASS 

Test Case 2

Input:

Enter a positive integer: 12345

--- MENU ---

1. Count even and odd digits
2. Sum of even and odd digits separately

Enter your choice: 2

Output:

Sum_even = 6

Sum_odd = 9

Result: PASS 

Test Case 3 — All Even Digits

Enter a positive integer: 2468

Enter your choice: 1

Output:

Even_dig = 4

Odd_dig = 0

Result: PASS 

Test Case 4 — All Odd Digits

Enter a positive integer: 13579

Enter your choice: 2

Output:

Sum_even = 0

Sum_odd = 25

Result: PASS 

```
(kali@kali)-[~/demo1]
└─$ cc even_odd_digit.c

(kali@kali)-[~/demo1]
└─$ ./a.out
Enter a positive integer: 12045

— MENU —
1. Count even and odd digits
2. Sum of even and odd digits separately
Enter your choice: 1

Even_dig = 3
Odd_dig = 2

(kali@kali)-[~/demo1]
└─$ ./a.out
Enter a positive integer: 12345

— MENU —
1. Count even and odd digits
2. Sum of even and odd digits separately
Enter your choice: 2

Sum_even = 6
Sum_odd = 9
```

7. Final Test Results

TC Number Choice Expected				Observed	Status
1	12045	1	3 even, 2 odd	3 even, 2 odd	✓ PASS
2	12345	2	Even sum 6, odd sum 9	Even sum 6, odd sum 9	✓ PASS
3	2468	1	4 even, 0 odd	4 even, 0 odd	✓ PASS
4	13579	1	0 even, 5 odd	0 even, 5 odd	✓ PASS
5	2468	2	Even sum 20, odd sum 0	Even sum 20, odd sum 0	✓ PASS
6	13579	2	Even sum 0, odd sum 25	Even sum 0, odd sum 25	✓ PASS
7	10020	1	4 even, 1 odd	4 even, 1 odd	✓ PASS
8	10020	2	Even sum 2, odd sum 1	Even sum 2, odd sum 1	✓ PASS
9	12345	3	Invalid choice	Invalid choice	✓ PASS
10	0	1	Invalid input	Invalid input	✓ PASS

Conclusion

The menu-driven C program was successfully compiled, executed, and tested. The switch-case correctly selects the required operation, while the program correctly identifies even and odd digits and calculates their separate sums. All prepared test cases produced the expected results.